Chapter 16

Adverse Drug Events, Medication Errors, and Drug Interactions

Vincent W.L. Tsui[1], Dixon Thomas[2], Shuhui Tian[3] and Allen J. Vaida[4]

[1]Queen Elizabeth Hospital, Hong Kong; [2]Gulf Medical University, Ajman, United Arab Emirates; [3]Beijing United Family Healthcare, Chaoyang District, Beijing, China; [4]Institute for Safe Medication Practices, Horsham, PA, United States

Learning Objectives:

Objective 16.1 Classify and detail adverse drug events.
Objective 16.2 Explain methods for prevention and management of adverse drug events.
Objective 16.3 Classify and detail medication errors.
Objective 16.4 Explain methods for prevention and management of medication errors.
Objective 16.5 Classify and detail drug interactions.
Objective 16.6 Explain methods for prevention and management of drug interactions.

OBJECTIVE 16.1. CLASSIFY AND DETAIL ADVERSE DRUG EVENTS

The World Health Organization (WHO) defines an adverse drug reaction (ADR) as "*a drug-related event that is noxious and unintended and occurs at doses used in humans for prophylaxis, diagnosis or therapy of disease or for the modification of physiological function*." Adverse drug event (ADE) is "*any untoward occurrence that may present during treatment with a pharmaceutical product but that does not necessarily have a causal relation to the treatment*." ADR has a causal relationship for the reaction to a drug. ADR is a type of ADE. Causes of an ADR could be directly attributed to a drug and its pharmacological properties. ADRs were always considered nonpreventable since they occur even if a drug is prescribed, dispensed, or administered appropriately. There is a growing understanding that ADRs may be preventable due to advances in our knowledge of how to prescribe and test drugs for certain patients (i.e., genomics). Many still consider these to preventable medication errors. ADEs that happen due to inappropriate drug use (medication error) or poor control of other contributing factors are considered preventable. ADEs are not necessarily caused by the drug's pharmacology itself. Insulin is known to cause hypoglycemia, but if it is rightly dosed, hypoglycemia is prevented. If the patient skips a meal one day without reducing the dose of insulin they may experience hypoglycemia. This is a preventable ADE occurring due to medication error of wrong drug administration. This incidence should not result in changes of the daily dose as long as meals and dose are scheduled accordingly. This concept highlights that healthcare professionals and consumers must communicate on proper drug administration and just not blame this occurrence as an ADR from the drug. There could be possibilities of many ADRs occurring even if the drug is appropriately prescribed, dispensed, and administered. A general confusion is prevalent in published articles to use ADEs and ADRs interchangeably.[1] Refer to Fig. 16.1. It compares the relationship between medication errors, ADEs, preventable ADEs, nonpreventable ADEs (ADRs), and potential ADEs.[32,33] ADEs have a significant cost both financially and in terms of quality of life. Few studies of ADRs have been conducted in the community, so the effect on primary care is harder to assess, but studies in the hospital environment have shown the following[1]:

- ADRs occur in 10%−20% of patients in the hospital
- ADRs are responsible for 5% of admissions to the hospital
- ADRs might be responsible for 1 in 1000 deaths in medical wards
- ADRs are the most common cause of iatrogenic injury in hospital patients

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00016-7
Copyright © 2019 Elsevier Inc. All rights reserved.

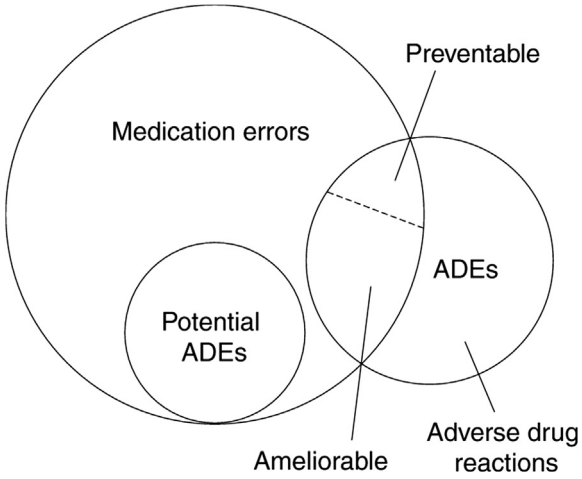


FIGURE 16.1 Relationship among medication errors, adverse drug events (ADEs), preventable ADEs, adverse drug reactions, and potential or preventable ADEs.[33] *Reproduced with permission from BMJ Quality & Safety.*

Classification of Adverse Drug Reactions

Some classification systems exist, but the most widely accepted is to group ADRs as types A to F (Table 16.1). Type A and type F are dose related and commonly occurring, while types B to E are uncommon.[2]

ADRs are also classified by severity into mild, moderate, severe, and lethal (Table 16.2).[4,5]

TABLE 16.1 Classification of Adverse Drug Reactions (ADRs)

Drug Reaction	Examples
Type A: Reactions occurring, given sufficient dose and duration of therapy: common and predictable (dose related or augmented)[2]	
Overdose (usually classified under adverse drug events)	• Hepatic failure with acetaminophen • Metabolic acidosis with aspirin
Side effects	• Nausea, headache with methylxanthines • Oral thrush or vaginal candidiasis with glucocorticoids • Nephrotoxicity with aminoglycosides
Secondary or indirect effects	• Diarrhea due to alteration in gastrointestinal bacteria after antibiotics • Phototoxicity with doxycycline or thiazide diuretics
Drug interactions	• Macrolide antibiotics increasing theophylline, digoxin, or statin blood levels
Type B: Drug hypersensitivity reactions restricted to a small subset of the general population: rare and mostly unpredictable (nondose related or bizarre)[2]	
Intolerance[a]	Tinnitus after a single aspirin tablet
Idiosyncrasy[b] (pharmacogenetics)	• Glucose-6-phosphate dehydrogenase deficiency: hemolytic anemia after oxidative drugs, e.g., dapsone[c] • Thiopurine methyltransferase deficiency: toxicity during azathioprine therapy[c] • Pseudoallergic reaction with nonsteroidal antiinflammatory drugs
Immunologic drug reactions (allergy)	• Anaphylaxis from beta-lactam antibiotics • Photoallergy with quinidine • Immune-mediated thrombocytopenia (with heparin) • Serum sickness (with antivenom preparations) • Vasculitis (with phenytoin) • Stevens–Johnson syndrome (with trimethoprim–sulfamethoxazole) • Drug-induced hypersensitivity syndrome (with allopurinol in HLA-B[a]58:01 individuals)[c]
Type C: Persisting for a relatively long time (dose related and time related or chronic)[3]	
"Continuing" reactions	Osteonecrosis of the jaw with bisphosphonates

Continued

TABLE 16.1 Classification of Adverse Drug Reactions (ADRs)—cont'd

Drug Reaction	Examples
Type D: Becoming apparent sometime after the use of a drug, which may make them more difficult to detect (time related or delayed)[3]	
"Delayed" reactions	Leucopenia with lomustine, which can occur up to 6 weeks after a dose
Type E: Associated with withdrawal of the drug (withdrawal or end of use)[3]	
"End-of-use" reactions	Insomnia, anxiety, and perceptual disturbances following the withdrawal of benzodiazepines
Type F: Unexpected failure of therapy (failure)	
Therapy failure	Therapy failure due to substandard or falsified drug products

[a]*Side effects at subtherapeutic doses.*
[b]*Drug effect not attributable to known pharmacologic properties of the drug and not immune mediated.*
[c]*This is an example of a type B reaction that is predictable.*

TABLE 16.2 Classification of Adverse Drug Reactions (ADRs) Based on Severity

Severity	Description	Example
Mild	No antidote or treatment is required; hospitalization is not prolonged.	• Antihistamines (some): drowsiness • Opioids: constipation
Moderate	A change in treatment, e.g., modified dosage, addition of a drug, but not necessarily discontinuation of the drug, is required; hospitalization may be prolonged, or specific treatment may be required.	• Hormonal contraceptives: venous thrombosis • Nonsteroidal antiinflammatory drugs: hypertension and edema
Severe	An ADR is potentially life-threatening and requires discontinuation of the drug and specific treatment of the ADR. May sometimes even cause disabilities, congenital anomaly, etc.	• ACE inhibitors: angioedema • Phenothiazines: abnormal heart rhythm
Lethal	An ADR directly or indirectly contributes to a patient's death.	• Nonsteroidal antiinflammatory drugs: Stevens–Johnson syndrome. • Desferrioxamine: congestive heart failure

OBJECTIVE 16.2. EXPLAIN METHODS FOR PREVENTION AND MANAGEMENT OF ADVERSE DRUG EVENTS

High-alert medication (also known as high-risk medication): Although most medications have a large margin of safety, a small number of drugs have a high risk of causing injury when they are misused. We call these "high-alert medications" to draw attention to this characteristic so that all involved in their use will treat them with the care and respect that they require. Errors may or may not be more common with these drugs than with the use of any others; however, the consequences of the errors are more devastating. For this reason, special considerations are required. These medications often need to be packaged differently, stored differently, prescribed differently, and administered differently than others. Whenever possible, "forcing function" methods that make it impossible for the drug to be given in a potentially lethal manner should be developed and instituted.[6]

Preventability of Adverse Drug Events

ADE is a negative health outcome. It is estimated that at least 60% of ADEs are preventable.[7] Several interventions have been used to prevent ADEs and can generally be categorized as provider- or system-based interventions. Provider- and system-based interventions should be used together to prevent ADEs optimally.

Provider-based approach—The individual clinician should ideally review the medication list at each patient encounter. Issues to consider range from dosage and directions to drug interactions and side effects.[8,9]

- **Avoid and be vigilant of high-alert drugs**—Several drug classes have been consistently associated with ADRs in studies of hospitalized patients.[10] Clinicians prescribing high-risk drugs should be vigilant of adverse effects and should consider discontinuing these drugs or replacing with drugs that are less likely to cause adverse effects.

- **Discontinue unnecessary drugs**—The risk of ADEs increases with the number of medications taken.[11–13] In addition to potentially harmful drugs, patients are often taking drugs that were previously given for a prior indication and are no longer needed.
- **Avoid drug–drug interactions**—The risk of an ADE due to drug–drug interactions is substantially higher when more medications are being prescribed. Most drug–drug interactions involve commonly used medications. As an example, the risk of bleeding with warfarin therapy is increased with coadministration of nonsteroidal antiinflammatory drugs (NSAIDs), selective serotonin reuptake inhibitors (SSRIs), omeprazole, lipid-lowering agents, and/or amiodarone.[14]
- **Adjust dosing based on age and creatinine clearance**—A common cause of ADEs is a failure to properly adjust doses for age and renal insufficiency. Adjusting appropriately for age is particularly important in pediatric and geriatric populations. Renal impairment can occur at any age but becomes more common with advancing age. For patients with stable renal function, creatinine clearance can be estimated according to published formulas that factor age into the calculation.[15]

System-based approach—Most ADEs are caused by poorly designed systems that either induce errors or make them difficult to detect. Although human error is usually the proximal event leading to an ADE, another fundamental fault is in the system that permits the error to reach the patient.[16]

High-leverage strategies fix the system; low-leverage strategies focus on the individual involved in an error. Because people cannot be expected to compensate for weak systems, error prevention tools that are designed to fix the system have a broader, more lasting impact (high leverage) than those directed at changing human behavior (low leverage).[17]

- **Computerized physician order entry (CPOE)**—CPOE refers to a variety of computer-based systems that facilitate the medication ordering process.[18] Some CPOE systems provide more than prescription-writing capabilities with fixed fields and drop-down menus for drug, dose, route, and frequency. These systems are further enhanced by the integration of clinical decision support systems (CDSSs) of varying sophistication. CDSS includes interactive software designed to assist health professionals in making clinical choices. CPOE with CDSS targets the prescribing stage of the medication use process and improves medication safety in the following ways:
  - Providing a means for standardization of practice
  - Improving the completeness and legibility of orders
  - Alerting clinicians to drug allergies, drug–drug interactions, and cumulative dose limits
  - Updating clinicians with the most current medication information
  - Providing dosage adjustment calculations based on patient characteristics
  - Timely communication of critical changes in a patient's condition, in turn facilitating appropriate adjustments

Several studies have found that CPOE with CDSS significantly prevents medication errors and ADEs. However, some others have reported unintended consequences related to the introduction of CPOE, especially among CPOE systems with limited decision support.[19–21]

- **Electronic medication administration record (eMAR)**—When CPOE has an electronic interface with an eMAR, the need for medication transcription is eliminated, removing errors caused by interpretation and translation. Moreover, eMAR helps organize medication administration schedules and prompts nurses to ensure timely medication administration.
- **Pharmacist interventions**—Pharmacists can play an important role in preventing ADEs through various interventions, such as patient counseling, medication reconciliation, and drug class-specific pharmacist services, e.g., anticoagulation clinics.
- **Others—Barcodes** can be affixed to medications and patient wristbands to ensure matching between patients, their eMAR, and their medications at the time of drug administration. **Smart pumps** are used to reduce errors associated with intravenous medication administration through their built-in safety features, such as safety alerts, clinical calculators, dose limits, and drug libraries. **Educational programs** may modestly decrease rates of medication errors and ADEs.[22] ADE reporting systems that include a subsection for ADRs should be established to encourage all professionals to report certain or probable/likely ADRs and other ADEs.

Use of Signals

Early identification of the hazards associated with drugs is the main goal of those involved in pharmacovigilance "signal detection," "signal generation," or "signaling." It refers to a process that aims to find, as soon as possible, any indication of an unexpected drug safety problem. A signal may indicate a new ADR or a change of the frequency of an ADR that is already known to be associated with the drugs involved.[23] Signals are not confirmation for the ADRs. Assessment of signals for a causal relationship between the drug and the reported ADE will confirm the ADRs. Signals are useful in taking measures to prevent and manage newer ADEs.[24]

Causality Assessment of Adverse Drug Reactions

Causality assessment is the method by which the extent of relationship between a drug and a suspected reaction is established, i.e., to attribute clinical events to drugs in individual patients or in case reports.[25] Many systems can be used to make a causality assessment of ADR reports among which the causality categories described by the Uppsala Monitoring Centre (Table 16.3) and the Naranjo algorithm (Tables 16.4 and 16.5) are the most commonly used.[26,27] Both of them have not been validated, and studies evaluating agreement between the two did not give consistent results.[28,29] The adoption of a particular system will be based on the consensus of the causality assessment committee involved. However, in case of analysis of serious ADRs, considerations of both systems should be used to ascertain causality.

TABLE 16.3 WHO–Uppsala Monitoring Centre Causality Categories

Term	Assessment Criteria
Certain	A clinical event, including laboratory test abnormality, occurring in a plausible time relationship to drug administration, and which cannot be explained by concurrent disease or other drugs or chemicals. The response to withdrawal of the drug (dechallenge) should be clinically plausible. The event must be definitive pharmacologically or phenomenologically, i.e., an objective and specific medical disorder or a recognized pharmacological phenomenon, using a satisfactory rechallenge procedure if necessary.
Probable/ likely	A clinical event, including laboratory test abnormality, with a reasonable time sequence to the administration of the drug. Unlikely to be attributed to concurrent disease or other drugs or chemicals, and which follows a clinically reasonable response to withdrawal (dechallenge). Rechallenge information is not required to fulfill this definition.
Possible	A clinical event, including laboratory test abnormality, with a reasonable time sequence to administration of the drug, but which could also be explained by concurrent disease or other drugs or chemicals. Information on drug withdrawal may be lacking or unclear.
Unlikely	A clinical event, including laboratory test abnormality, with a temporal relationship to drug administration, which makes a causal relationship improbable but not impossible, and in which other drugs, chemicals, or underlying disease provide plausible explanations.
Conditional/ unclassified	A clinical event, including laboratory test abnormality, reported as an adverse reaction, about which more data are essential for a proper assessment, or the additional data are under examination.
Unassessable/ unclassified	A report suggesting an adverse reaction that cannot be judged because information is insufficient or contradictory, and which cannot be supplemented or verified.

TABLE 16.4 Naranjo Algorithm: Adverse Drug Reaction Probability Scale

Questions	Yes	No	Do Not Know
1. Are there previous conclusive reports on this reaction?	+1	0	0
2. Did the adverse event appear after the suspected drug was administered?	+2	−1	0
3. Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	+1	0	0
4. Did the adverse reaction reappear when the drug was readministered?	+2	−1	0
5. Are there alternative causes (other than the drug) that could on their own have caused the reaction?	−1	+2	0
6. Did the reaction reappear when a placebo was given?	−1	+1	0
7. Was the drug detected in the blood (or other fluids) in concentrations known to be toxic?	+1	0	0
8. Was the reaction more severe when the dose was increased, or less severe when the dose was decreased?	+1	0	0
9. Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	+1	0	0
10. Was the adverse event confirmed by any objective evidence?	+1	0	0

TABLE 16.5 Naranjo Algorithm: Adverse Drug Reaction Probability Categories

Score	Category
≥9	Definite
5–8	Probable
1–4	Possible
0	Doubtful

OBJECTIVE 16.3. CLASSIFY AND DETAIL MEDICATION ERRORS

The National Coordinating Council for Medication Error Reporting and Prevention (NCCMERP) defines medication errors as those preventable events that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of the healthcare professional, patient, or consumer. Such events may be related to professional practice, healthcare products, procedures, and systems, including prescribing, order communication, product labeling, packaging, nomenclature, compounding, dispensing, distribution, administration, education, monitoring, and use.[30]

In simple terms, the error is a nondeliberate deviation from what is intended, whereas violation is a deliberate deviation. An action that does not go as intended is a so-called error of execution and may be further described as being either a slip, if the action is observable, or a lapse, if it is not. An example of a slip is accidentally pushing the wrong button on a piece of equipment. An example of a lapse is a memory failure, such as forgetting to administer a medication. The intended action is the wrong one: a failure that occurs when the intended action is actually incorrect is clearly a "mistake."[31]

- *Slip*: Drive to the store and head to the office. You know where the store is but are in automatic mode to go to work. You figure a dose of 20 mg/kg as 1400 mg in a 70 kg person but write 140 mg on the order.

A mistake is a failure of planning. This can be either rule based, when the wrong rule is applied, or knowledge based, when a clinician does not take the correct course of action. An example of a rule-based mistake would be getting the diagnosis wrong and so embarking on an inappropriate treatment plan. Knowledge-based mistakes tend to occur when providers are confronted with unfamiliar clinical situations.[31]

- *Mistake*: Drive to the store but go the wrong way because you are not sure of where it is or make the wrong turn. You had a dose of 20 mg/kg and calculated to report 140 mg dose for a 70 kg person.

Root cause analysis is a highly structured systemic approach to incident analysis that is generally reserved for the most serious patient harm episodes.[31]

Types of Medication Errors[32,33]

Medication errors can occur from the clinician who prescribes the medication to the healthcare professional who administers the medication or the patient who self-administers the medication.

Different types of medication errors include (but are not necessarily limited to) the following:

- Prescribing errors
- Omission errors
- Wrong time errors
- Improper dosing errors
- Wrong dose errors
- Improper administration technique errors
- Wrong drug preparation errors
- Fragmented care errors

The Scope of the Problem

In December 1999, the Institute of Medicine (IOM) now called the National Academy of Medicine (NAM) released a report, *To Err Is Human*, that identified medical error as a significant public health problem claiming between 44,000 and

98,000 deaths each year. Medication errors alone were estimated to account for over 7000 deaths annually. This report compellingly summarized the staggering nature and extent of error in healthcare and convincingly thrust medication safety into the forefront as a national priority.[34]

The release in 2006 of another IOM report, *Preventing Medication Errors*, suggested that all stakeholders, including accreditation agencies, state professional boards, and relevant state and federal agencies, should promote medication error reporting more aggressively.[35] Two other studies, namely a 2006 report by the Centers for Disease Control and Prevention and a follow-up in 2016, estimated that 2.4 emergency department visits per 1000 population are due to ADEs and this rate increased to 4 visits per 1000 population in almost 10 years.[36,37]

The medication use process is complex, with many steps, handoffs, and opportunities for error. From ordering, to dispensing, to administering a medication, we must continuously assess the design of systems and processes to refine what we do, how we do it, and the environment in which the work is carried out. In short, preventing errors requires better system design and innovative tools to support safe systems. Much is known about the causes and prevention of medication errors. Yet, our collective challenge to turn this knowledge into practice remains unmet, and harmful medication errors represent a serious concern throughout the world. Other published studies have proven that medication error is a global concern. A systematic review that analyzed 45 studies from 10 middle eastern countries reported the incidence of medication errors across the medication use process.[38] Another systematic review that analyzed 17 studies from six Southeast Asian countries identified medication administration to be the most reported type of medication error (15.2% −88.6%).[39] An Irish study showed that medication error incidents in the acute care setting led to patient harm in 66.6% of the cases.[40] In 2017, WHO launched its third global patient safety challenge, *Medication Without Harm*. WHO has called on all countries and key stakeholders to make strong commitments, prioritize and take early action, and effectively manage three key areas to protect patients from harm, namely high-risk situations, polypharmacy, and transitions of care.[41] In a follow-up article, reminiscent of the call to action from the IOM 2006 report on medication errors, the authors again called on politicians and healthcare leaders to prioritize medication safety. Implementing measures to reduce harm will need to include educating and empowering patients and carers, developing tools to assist frontline healthcare professionals, and engineering new systems of care to create resilience against factors that predispose patients to the risk of medication errors.[42,43]

Despite these calls to action almost two decades ago, healthcare is still struggling to address medication safety across the continuum of care. There exists much information on medication errors as well as prevention methods and strategies, but unfortunately many of these known recommendations are never shared.

Categorization of Medication Errors

The NCCMERP categorizes the severity of error outcomes on a scale known as the medication error index.[44] It considers factors such as whether the error reached the patient and to what degree the patient was harmed. The categories range from A through I (Table 16.6).

TABLE 16.6 Medication Error Index

No Error
(A) Circumstances or events that have the capacity to cause error
Error, No Harm
(B) An error occurred but the medication did not reach the patient **(C)** An error occurred that reached the patient but did not cause patient harm **(D)** An error occurred that resulted in the need for increased patient monitoring but no patient harm
Error, Harm
(E) An error occurred that resulted in the need for treatment or intervention and caused temporary patient harm **(F)** An error occurred that resulted in initial or prolonged hospitalization and caused temporary patient harm **(G)** An error occurred that resulted in permanent patient harm **(H)** An error occurred that resulted in a near-death event, e.g., anaphylaxis, cardiac arrest
Error, Death
(I) An error occurred that resulted in patient death

OBJECTIVE 16.4. EXPLAIN METHODS FOR PREVENTION AND MANAGEMENT OF MEDICATION ERRORS

Common Causes of Medication Errors

The causes of medication errors are often multifactorial, ranging from human lapses and system loopholes to external factors. The common reasons are as follows[45]:

- Mistakes in writing or filling prescriptions
- Failure to recognize drug interaction
- Failure to properly instruct patient on drug dosage and administration
- Wrong dosage administered
- Similar drug names or medication appearance
- Medical abbreviations
- Failure to properly input prescription onto a computer database
- Poorly designed systems that either induce errors or make them difficult to detect
- Problems with ward-based equipment, e.g., with access or functionality
- Problems with drug supply and storage, e.g., pharmacy dispensing errors or issues with ward stock management
- Staff health status, e.g., fatigue, stress
- Interruptions or distractions during drug administration

Measures to Be Taken to Prevent Medication Errors[46]

Various methods and strategies have been suggested to prevent medications errors, involving both patients and clinical staff (Table 16.7).

Some other measures can be taken to decrease those medication errors,[47,48] such as:

- electronic prescribing and CPOE with CDSSs
- medication error-reporting systems
- alerts about medication errors
- smart infusion pumps
- telemedicine, telehealth, or telepharmacy
- computerized medication box

The Institute for Safe Medication Practices—Sharing and Learning From Errors

The Institute for Safe Medication Practices (ISMP) was formally established in the United States as a nonprofit organization in 1994. However, its reporting program is among the earliest and most aggressive in the battle to prevent medication errors. It also provided error prevention information and encouraged practitioners to submit reports of medication errors so that further recommendations for prevention could be published. Today, collection and analysis of error

TABLE 16.7 Possible Actions by Patients and Clinical Staff to Prevent Medication Errors

Patients	Clinical Staffs
• Find out the name of each of your medications	• Adhere to the "five rights" of medication safety, i.e., administering the right medication, in the right dose, at the right time, by the right route, to the right patient
• Ask questions about how to use the medications	
• Know what your medications are for	
• Read drug labels and follow directions	• Do provide appropriate measurable devices if needed
• Keep all of your healthcare providers informed about your medications and dietary supplements, including vitamins and herbals	• Conduct medication reconciliation with any transition in care, including allergy information and adverse drug reactions
• Keep a list of your medications with you at all times and let your next of kin know	• Consider interventions to reduce medication errors

information from the ISMP National Medication Errors Reporting Program (MERP), Food and Drug Administration (FDA) MedWatch, and many other sources around the world are the foundation on which ISMP's collaborative, educational efforts are based.

ISMP operates three national, voluntary, medication error-reporting programs, namely the ISMP National MERP, the ISMP National Vaccine Errors Reporting Program, and the ISMP National Consumer MERP.

These voluntary reporting programs differ from mandatory external reporting programs in several ways. The reports come directly from frontline practitioners and consumers rather than risk, quality, or safety managers. The narrative descriptions (and pictures) that accompany the reports offer rich content for learning compared with mandatory reports that often prompt for standard information and a succinct description of the event, but not its causes. Communication with the practitioners or consumers who were involved in or detected the event is possible and potentially more helpful than communication with managers screening and completing the mandatory reports. The reports to ISMP are sent for altruistic reasons—because practitioners want to share information that will benefit others, not because there is a mandate to do so.

In November 2006, the International Medication Safety Network was launched. This is a group of safe medication practice centers, including members from the United States and WHO, committed to prevent medication errors and to contribute to safer care. The participants pledged to work together to promote achievement of these essential objectives to encourage and further the development of safe medication practice centers in all countries and to facilitate cooperation among them.[49]

Error-reporting systems represent one of the primary means by which healthcare providers can learn about

- *close calls (near misses)*—errors that occurred but were caught before they reached a patient
- *potential risks*—hazardous conditions hidden in processes
- *actual errors*—errors that occur during the delivery of patient care
- *causes of errors*—underlying weaknesses in systems and processes that explain why an error happened
- *error prevention*—ways to prevent recurrent events and, ultimately, patient harm

Error reporting is a fundamental component of a safety culture, but persuading healthcare workers to submit reports is no easy task given the potential disincentives to reporting. First, people's reactions to making errors are variable, but candid confessions of mistakes are not particularly popular. In fact, people have a natural desire to forget that the incident ever happened. Even if workers are willing to speak up about errors, they may still believe that the extra work is not worth their time if they feel that no good will come from reporting. The workforce is also understandably reluctant to report errors if they are worried that the information will get them or their colleagues in trouble, legally or socially, or impact their job.

It is important to report errors externally as well as internally. External reporting can lead to improved safety in several ways.[50]

- Alerts about new hazards (e.g., complications of a new drug) can be generated from even a few reports.
- Information about the experience of individual hospitals in using new methods to prevent errors can be disseminated.
- Central analysis of many reports can reveal trends and hazards that require attention.
- Central analysis can lead to recommended "best practices" for all to follow.

Many best practices for error prevention exist and are included in ISMP's publications, US FDA warning and alerts, accreditation agencies standards, and in published literature. ISMP has a list of high-alert medications, which are drugs that bear a heightened risk of causing significant patient harm when used in error.[6]

Strategies to prevent errors with high-alert medications include standardizing drug concentrations, controlled storage, automated alerts for prescribing, and the use of checklists and independent double checks.

Using tall man or mixed case lettering and including indications when prescribing medications are strategies to help prevent medication that may look or sound alike.[51,52]

Because medication strengths, e.g., mg, mcg, and doses, e.g., mg/kg, are in metric units, using metric units of measure to express the volume of liquid medications, weigh patients, and document the weight on medical records and prescriptions is another recommendation.

The use of technology to help prevent errors should be employed, such as barcode scanning of medications during the dispensing process and during drug administration; electronic prescribing; workflow system to prepare parenteral medications; infusion devices with smart pump technology; and robotics for storage, dispensing, and preparation of medications.

Finally, having pharmacists available on patient care units in hospitals and counseling patients on their medications in the outpatient setting has shown to prevent errors as well as patient harm.

Although the work of medication safety is the responsibility of many, the coordination of the efforts and assuring communication across all professionals is optimized if the effort stems from one individual. This role is being served with

the position of a medication safety officer (MSO). Healthcare organizations, both acute and ambulatory, throughout the world, are adding this position. Either under the direction of an MSO or others, each organization should develop a strategic plan for medication safety and include the importance of sharing and learning from medication errors.

OBJECTIVE 16.5. CLASSIFY AND DETAIL DRUG INTERACTIONS

A drug interaction is said to occur when an agent (the precipitant agent) alters the effect of a drug (the object drug). Drug interactions are generally undesirable because they can result in a decrease in efficacy or increase in toxicity. When doxycycline is taken with iron sulfate, it forms poorly soluble chelates with iron, thus decreasing the absorption of both preparations and the concentration of doxycycline at the site of action. When allopurinol and mercaptopurine are used together, allopurinol inhibits the metabolism of mercaptopurine by xanthine oxidase, mercaptopurine accumulates, and its toxic effects, such as leucopenia and thrombocytopenia, are more likely to develop.

Desirable Drug Interactions

There are also situations in which drug interactions are beneficial, e.g., caffeine, a coanalgesic found in many paracetamol preparations, is included to increase the absorption rate of paracetamol through unclear mechanisms. Contrary to undesirable ones, these drug pairs interact to increase efficacy and/or decrease toxicity (Table 6.8) through various mechanisms.

Drug Interactions Precipitated by Agents Other Than Drugs

Although drug interactions usually involve two drugs, more specifically called drug−drug interactions, the precipitant agent is not always a drug (Table 16.9), but sometimes a herb, food, alcohol, or cigarettes. It would require awareness of social history to recognize them and a holistic approach in counseling to manage them, like diet control and smoking cessation.

Some drug pairs interact in unique ways but much more in a similar manner. Drug interactions are typically classified by mechanism into pharmaceutical, pharmacokinetic, and pharmacodynamic drug interactions.

TABLE 16.8 Examples of Desirable Drug Interactions[53]

Category	Example
Increase Efficacy	
Synergistic	Bactericidal activity in enterococcal infective endocarditis is achieved by the combination of a penicillin, e.g., ampicillin, and an aminoglycoside, e.g., gentamicin, that either alone only inhibits bacterial growth
	Rifampicin, isoniazid, and other antibiotics are used in combination in the treatment of tuberculosis to lower the chance of resistance despite the risk of increased hepatotoxicity
Augmentative	Because of extensive cytochrome P450 (CYP) 3A4 metabolism, lopinavir alone has an exceedingly low plasma concentration so that commercially it is only available in combination with low-dose ritonavir, which inhibits CYP3A4 and increases exposure to lopinavir
Facilitative	Beta-lactamase inhibitors, e.g., clavulanic acid, are used in combination with beta-lactam antibiotics, e.g., amoxicillin, to improve coverage in beta-lactamase producing bacteria
Decrease Toxicity	
Reparative	Constipating aluminum hydroxide is used with laxating magnesium hydroxide in antacids to cancel out the adverse effects of one another
Antidotal	Folic acid is taken, usually at least 24 h after weekly methotrexate, to decrease the gastrointestinal and mucosal adverse effects
Increase Efficacy and Decrease Toxicity	
Complementary	Combinations of thiazide diuretics, loop diuretics, and potassium-sparing diuretics are used to treat refractory edema, lower the doses of individual diuretics, and/or avoid the use of potassium supplements
	Carbidopa is used together with levodopa to inhibit the peripheral conversion of levodopa to dopamine by DOPA decarboxylase, thus minimizing peripheral dopaminergic adverse effects and reserving more levodopa for the central nervous system.

TABLE 16.9 Examples of Drug Interactions With Nondrug Precipitant Agents

Precipitant Agent	Example
Herb	St. John's wort may increase the metabolism of warfarin by inducing CYP2C9 and CYP3A4
Food	Grapefruit juice may decrease the metabolism of carbamazepine by inhibiting CYP3A4
	Tyramine-rich foods may cause a hypertensive crisis in patients taking monoamine oxidase inhibitor
Alcohol	Alcohol may add to the CNS depressant effect of benzodiazepines
	Alcohol may precipitate flushing, tachycardia, breathlessness, and hypotension, called the disulfiram reaction, in patients taking metronidazole
Cigarettes	Smoking may increase the metabolism of theophylline by inhibiting CYP1A2

TABLE 16.10 Examples of Pharmaceutical Drug Interactions

Precipitant Agent	Example
Another drug	Ketorolac is alkaline in solution and incompatible if mixed with an acidic drug, e.g., haloperidol, in continuous subcutaneous infusion[54]
Diluent	Amphotericin B may precipitate with sodium chloride solution
	Ceftriaxone may precipitate with calcium-containing fluids
Container or infusion set	Carboplatin forms precipitate with needles or sets with aluminum parts
	Polysorbate 80 in amiodarone injection may leach di-2-ethylhexyl phthalate from polyvinyl chloride administration sets

Pharmaceutical Drug Interactions (Incompatibilities)

In pharmaceutical drug interactions, also known as pharmaceutical incompatibilities, the precipitant agent affects the object drug before administration. They usually represent physical incompatibilities in parenteral routes and the precipitant agent can be another drug, the diluent, the container, or the infusion set (Table 16.10).

Pharmacokinetic Drug Interactions

In pharmacokinetic drug interactions, the precipitant agent alters the absorption, distribution, metabolism, and/or excretion of the object drug (Table 16.11).

Many drug interactions are attributed to more than one mechanism. For example, it is recognized that the use of salicylates or NSAIDs is a risk factor for methotrexate toxicity.[56] A few mechanisms have been proposed: decrease in renal perfusion, competition for tubular excretion, and displacement from plasma proteins. The significance of each mechanism is not always determined.

Clinically significant protein-binding interactions are particularly uncommon. Taking the example of phenylbutazone and warfarin, while phenylbutazone effectively displaces warfarin from plasma proteins, warfarin has a low extraction ratio in the kidney, which means the body has much capacity to eliminate the free drug displaced. In turn, the interaction is attributed to other mechanisms, e.g., that phenylbutazone inhibits the metabolism of *S*-warfarin, the more potent enantiomer,[57] or that phenylbutazone increases the risk of gastrointestinal bleeding as an NSAID.

Pharmacodynamic Drug Interactions

In pharmacodynamic drug interactions, the precipitant agent alters the dose—response relationship of the object drug with an agonistic or antagonistic effect (Table 16.12). Sometimes two drugs do not directly alter the effect of one another, but the simple addition of their effects warrants caution, for example, the concurrent use of two drugs that prolong the QT

TABLE 16.11 Examples of Pharmacokinetic Drug Interactions[55]

Subcategory	Common Mechanism	Example
Absorption	Adsorption or chelation	Cholestyramine decreases the absorption of warfarin by complexation
	Changes in gastrointestinal pH	Pantoprazole decreases the absorption of ketoconazole by increasing the gastric pH and decreasing the solubility of ketoconazole
	Induction or inhibition of transporter proteins	Rifampicin decreases the bioavailability of dabigatran by inducing P-glycoprotein and increasing the ejection of absorbed dabigatran
	Changes in gastrointestinal motility	Propantheline decreases the absorption rate of paracetamol in the small intestine by slowing gastric emptying
Distribution	Changes in plasma protein binding	Phenylbutazone increases the concentration of free warfarin by displacing warfarin from plasma proteins
	Induction or inhibition of transporter proteins	Ketoconazole increases the concentration of ritonavir in the cerebrospinal fluid (CSF) by inhibiting P-glycoprotein and decreasing the ejection of ritonavir from the CSF
Metabolism	Enzyme induction	Phenytoin increases the metabolism of quetiapine by inducing CYP3A4
	Enzyme inhibition	Paroxetine decreases the metabolism of tamoxifen to one of its more active metabolites by inhibiting CYP2D6
Excretion	Changes in renal blood flow	Mefenamic acid decreases the urinary excretion of lithium by inhibiting renal prostaglandin synthesis and decreasing renal blood flow
	Changes in urinary pH	Sodium bicarbonate increases the urinary excretion of methotrexate by alkalinizing the urine and increasing the solubility of methotrexate
	Changes in tubular excretion	Probenecid decreases the urinary excretion of ciprofloxacin by competing for tubular excretion

TABLE 16.12 Examples of Pharmacodynamic Drug Interactions

Subcategory	Level	Example
Additive	Receptor-level	Both chlorpheniramine and atropine exhibit anticholinergic properties and may worsen adverse effects such as blurred vision, dry mouth, and constipation
	Effect-level	Both opioids and benzodiazepines can cause drowsiness, and caution is advised with concurrent use
Synergistic	Receptor-level	Amitriptyline inhibits the reuptake of noradrenaline into adrenergic neurons and increases the response to adrenaline
	Effect-level	Thyroxine corrects hypothyroidism and increases the metabolism of clotting factors, thus decreasing the requirement of warfarin
Antagonistic	Receptor-level	Haloperidol blocks dopamine receptors in the brain and antagonizes the effect of levodopa
	Effect-level	Hydrocortisone increases the blood glucose level and diminishes the therapeutic effect of gliclazide

interval, such as citalopram and domperidone. Whether the effects of two drugs are additive, more than additive, or less than additive requires extensive research to establish, but the important point in practice is to recognize the risk of potential synergism in agonistic drug interactions.

Interindividual Variations

Understanding the mechanisms is an essential yet only the first step to manage a drug interaction. It is equally important to recognize that the clinical manifestations are highly variable in individual patients. Some CYP isoenzymes, e.g., 2C9 and

2C19, and some transporter proteins, e.g., P-glycoprotein, are subject to genetic polymorphism, which means variations in the expression of these proteins and therefore pharmacokinetic interactions. As for pharmacodynamic interactions, some outcomes are measurable, e.g., the QT interval, and some are not, like drowsiness. They will require more observation and communication to evaluate. Healthcare professionals must keep these factors in mind when approaching a drug interaction.

OBJECTIVE 16.6. EXPLAIN METHODS FOR PREVENTION AND MANAGEMENT OF DRUG INTERACTIONS

Identification of drug interactions used to be a painful manual process: matching the many drugs on a prescription to the many entries in a reference book. General references, e.g., the *British National Formulary*, list drug interactions under the monographs of individual drugs, whereas more specialized references, like *Stockley's Drug Interactions*, provide monographs of drug pairs. With the help of information technology, drug interaction checkers are available on the Internet and subsequently as applications on handheld devices. Regardless of the tools used, however, some principles are consistently applied:

- Be vigilant with potent enzyme inducers, e.g., carbamazepine, phenytoin, and rifampicin, and potent enzyme inhibitors, e.g., azoles and protease inhibitors (Table 16.13). They can affect more than one isoenzyme and altogether a wide range of drugs to various extents, thus making them common culprits of drug interactions. In particular, carbamazepine, known as an autoinducer, induces its own metabolism. Its half-life starts to decrease 3−5 days after initiation; maximal induction occurs in 3 weeks, and autoinduction may take up to 5 weeks to complete.[58] Coupled with hetero-induction by phenytoin, which takes 1−2 weeks, and that by phenobarbital, which takes 2−3 weeks, the use and therefore dose titration of any concurrent CYP3A4 substrates require extreme caution.

TABLE 16.13 Examples of Potent (Drug Names in Bold) and Moderate Enzyme Inducers and Inhibitors[59]

Isoenzyme	Inhibitors	Inducers
CYP1A2	**Fluvoxamine** Ciprofloxacin Combined hormonal contraceptives	Phenytoin
CYP2C9	Amiodarone Fluconazole 200 mg daily Miconazole	Rifampicin
CYP2C19	**Fluconazole 100 mg daily** **Fluvoxamine** Omeprazole 40 mg daily Voriconazole	Rifampicin
CYP2D6	**Bupropion** **Fluoxetine** **Paroxetine** **Quinidine** **Terbinafine** Duloxetine	
CYP3A4	**Clarithromycin** **Protease inhibitors** **Itraconazole** **Ketoconazole** **Voriconazole** Diltiazem Dronedarone Erythromycin Fluconazole Posaconazole Verapamil	**Carbamazepine** **Phenytoin** **Rifampicin** Bosentan Efavirenz

- Pay attention to drugs with a narrow therapeutic window or requiring specific levels to work, e.g., anticoagulants, antiepileptics, antimicrobials, and immunosuppressants. It is not always easy to put one drug interaction before another, but these drugs are often associated with more pressing concerns.
- Prioritize drug interactions with permanent or detrimental consequences, e.g., organ damage and contraceptive failure, for example; clarithromycin inhibits the metabolism of ergotamine by CYP3A4, and concurrent use of the two is contraindicated because of a high risk of limb ischemia and ergotism.
- Evaluate the evidence of drug interactions. Like that of the efficacy and safety of a drug, the evidence of a drug interaction ranges from extensive research to case reports, from in vitro studies to theories. Although stronger evidence points to the certainty of a drug interaction, weaker evidence gives us insights into rarer ones.
- Always consider patient-specific factors, including liver and kidney function, as well as the history of drug allergies and ADRs. They help to identify patients at risk and determine the feasibility of alternatives.

Risk Ratings of Drug Interactions

When a drug interaction is identified, most practitioners consider its clinical significance next. Unlike the pregnancy risk categories and the common terminology criteria for adverse events in the United States, there is no one single widely accepted grading scale for drug interactions. Many clinical resources, e.g., *Lexicomp* and *Medscape*, have developed their systems of risk ratings (Table 16.14). Each of them involves four to five ratings that indicate the clinical significance of drug interaction and the corresponding action level from modification to no action. Some clinical resources have also assigned ratings to evidence, from extensive to theoretical, and onset, from immediate to delayed.

On top of any one single rating, there is a wide spectrum of strategies to manage a drug interaction. In the example of combined oral contraceptives with phenytoin, contraception is less reliable because phenytoin increases the metabolism of the contraceptive steroids and may allow ovulation to occur. Considering the serious consequences of an unplanned pregnancy, as well as fetal malformation due to antiepileptics with reproductive effects, every possible step should be taken to reduce the risks. Switching to an unaffected contraceptive method, using additional contraceptive precautions, shortening hormone-free interval of 3−4 days, maintaining an ethinylestradiol dose of at least 50 μg daily, and switching to a noninteracting antiepileptic are all possible.[59] Although the luxury of having multiple options is not a must, this scenario has certainly highlighted the many dimensions one can have in managing a drug interaction.

Discontinuation

The first step to prevent and potentially manage a drug interaction is to assess the necessity of drugs involved, which also forms an integral part of a pharmacist's role. For example, if pantoprazole decreases the absorption of ketoconazole, it is simple yet often overlooked to check whether pantoprazole is indicated. Discontinuing it or switching to an H_2-receptor antagonist when necessary would be a better solution than separating the doses of the two drugs considering pantoprazole's long duration of action. In the case that NSAID increases the risk of methotrexate toxicity, avoiding the use of NSAIDs is probably the safest way of management.

Therapy Modification

In the case that both drugs are indicated, the rest is a balance of modifying and monitoring. On the one end of the balance, there is a wide range of modifications to therapy that may manage a drug interaction, including withdrawing one or more

TABLE 16.14 Examples of Risk Rating Systems for Drug Interactions Developed by Clinical Resources

Lexicomp	Medscape	Epocrates	Micromedex	Stockley's
X: Avoid combination	Contraindicated	Contraindicated	Contraindicated	Avoid
D: Consider therapy modification	Serious—use alternative	Avoid/use alternative	Major	Adjust
C: Monitor therapy	Monitor closely	Monitor/modify treatment	Moderate	Monitor
B: No action needed	Minor	Caution advised	Minor	Informative
A: No known interaction			Unknown	No action

TABLE 16.15 Examples of Recommendations on Dose Modification for Drug Interactions

Drug Interaction	Recommendation
Diltiazem decreases the metabolism of simvastatin by inhibiting CYP3A4	Restrict the maximum dose of simvastatin to 20 mg daily[60]
Allopurinol decreases the metabolism of mercaptopurine by inhibiting xanthine oxidase	Decrease the dose of mercaptopurine by 75%[61]
Rifampicin increases the metabolism of lapatinib by inducing CYP3A4	Titrate the dose of lapatinib from 1250 mg daily up to 4500 mg daily or from 1500 mg daily up to 5500 mg daily[62]

drugs, switching to a noninteracting or less-interacting alternative, and adjusting the dosage of the drugs. Clear recommendations from authorities or manufacturers on dose modification may be available for drug interactions that are more significant or better studied (Table 16.15).

Monitoring Without Interventions

On the other end, monitoring is the mainstay of management when there are no alternatives, e.g., monitor the QT interval if two QT-prolonging agents must be used together. It is also possible to decide the course of action through monitoring, like decreasing the dose of lithium with concurrent NSAIDs as guided by the lithium level or decreasing the dose of warfarin with concurrent thyroxine by checking the international normalized ratio. In between the two ends, every factor is considered on a case-by-case basis.

Balance of Strategies

In an example of drug interaction between simvastatin and clarithromycin, simvastatin, an HMG-CoA reductase inhibitor used in the treatment of dyslipidemia, is extensively metabolized by CYP3A4. Clarithromycin, a macrolide, inhibits CYP3A4 and increases the concentration of simvastatin, therefore the risk of toxicity, e.g., myopathy and rhabdomyolysis. Switching to another statin or another macrolide does not appear to eliminate the interaction, not to mention that another macrolide may not be the most appropriate antimicrobial. In this case, given the usual short-term use of antimicrobial therapy, it is prudent to temporarily withdraw simvastatin during the course of clarithromycin.

In another example of drug interaction between tamoxifen and paroxetine, tamoxifen, a selective estrogen receptor modulator used in the treatment of breast cancer, is metabolized by CYP2D6 to antiestrogens that are more active. Paroxetine, an SSRI, inhibits CYP2D6 and may decrease the efficacy of tamoxifen. In reality, it is difficult to ascertain or monitor its impact on the long-term risk of breast cancer recurrence. Given the potential seriousness of this interaction, modifying either drug is a more desirable strategy. As for paroxetine, it is possible to switch to a weak CYP2D6-inhibiting SSRI, such as citalopram and escitalopram, or a non-SSRI antidepressant, e.g., venlafaxine. If the patient does not respond to other antidepressants, the possibility of switching tamoxifen to an aromatase inhibitor can be explored in postmenopausal patients.

In the two examples above, the plan of management would have been very different if it is known, for example, that paroxetine is expected to complete in 2 weeks, or that clarithromycin is used for its long-term immunomodulatory effects. Even when a drug interaction has a standard or straightforward plan of management, there are a few points that every pharmacist must bear in mind:

1. Pay attention to long-term effects when withdrawing one or more drugs, e.g., SSRIs. Always distinguish between temporary and permanent withdrawal. Watch out for unintended discrepancies with any transition in care.
2. Beware of introducing new interactions when switching to alternatives, for example, switching a proton pump inhibitor to an antacid may lessen its impact on the gastric pH but may introduce adsorption or complexation with other drugs.
3. Communicate any changes with patients and carers as far as practicable. They must understand which drug to stop, which one to start, and what doses to take to effect any changes. Often monitoring of signs and symptoms goes beyond laboratory results and requires a prompt response from patients and carers in case of emergency, e.g., when signs of hypoglycemia appear after titrating the doses of antidiabetic drugs.

CONCLUSION

The role of pharmacists in preventing and managing drug-related problems is emphasized. Pharmacotherapy knowledge, in which pharmacists are experts, provides the foundation for classifying and understanding these problems, whereas clinical pharmacy skills give us the tool to prevent and manage them, with a patient-centered approach. Many of the risks and measures are not absolute and present an opportunity for every pharmacist to contribute.